# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 21 | Square first, then multiply. |
| 2 | **(a)** 50 | Evaluate each bracket then multiply. |
| 3 | **(c)** $16x + 6$ | Perimeter $= 2(l + b)$. |
| 4 | **(b)** 55 | Triangular numbers. |
| 5 | **(c)** $2n + 4$ | Check by substituting a value. |
| 6 | **(b)** 22 | Substitute carefully; xy is a product. |
| 7 | **(d)** 23 | Apply the rule. |
| 8 | **(d)** 13 | Square first, then multiply. |
| 9 | **(d)** 70 | Evaluate each bracket then multiply. |
| 10 | **(a)** $18x + 6$ | Perimeter $= 2(l + b)$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Expand: $7x - 14 - 3x + 15$ (1); $4x + 1$ (1). | 1+1 |
| 12 | $2n + 2(n + 1) + 2(n + 2)$ (1); $= 6n + 6 = 6(n + 1)$ (1). | 1+1 |
| 13 | Substitute: $2(5)^2 - 3(5) + 4$ (1); $= 39$ (1). | 1+1 |
| 14 | Expand: $9x - 27 - 4x + 8$ (1); $5x - 19$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $P = 2[(5x + 4) + (3x - 1)]$; $= 2[8x + 3]$ (1); $= 16x + 6$ (1); At $x = 6$: 102 cm (1). | 1+1+1 |
| 16 | $4n$ is even (even × any integer); Even + odd = odd (1); $n = 5$: 23 odd (1); $n = 6$: 27 odd (1). | 1+1+1 |
| 17 | A $= 20 + 15k$, B $= 60 + 10k$; Equate: $5k = 40$ (1); $k = 8$ km (1); Fare then ₹140 (1). | 1+1+1 |
| | **OR:** $P = 2[(7x + 3) + (2x - 1)]$; $= 2[9x + 2]$ (1); $= 18x + 4$ (1); At $x = 5$: 94 cm (1). | 1+1+1 |
| 18 | $6n$ is even (even × any integer); Even + odd = odd (1); $n = 7$: 43 odd (1); $n = 8$: 49 odd (1). | 1+1+1 |
| | **OR:** A $= 30 + 12k$, B $= 90 + 7k$; Equate: $5k = 60$ (1); $k = 12$ km (1); Fare then ₹174 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $4(15) + 3 = 63$ (1); (ii) $4n + 3 = 91$ gives $4n = 88$ (1); $n = 22$ (1); (iii) Figure 1 = 7, figure 2 = 11 (1); Total = 18 (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 15$ (1); Total $= x + 2x + 15 = 3x + 15$ (1); $3x + 15 = 135$ so $3x = 120$ (1); $x = 40$ (1); Brother = ₹95 (1). | 1+1+1+1+1 |
| 20 | (i) $5(12) + 2 = 62$ (1); (ii) $5n + 2 = 92$ gives $5n = 90$ (1); $n = 18$ (1); (iii) Figure 1 = 7, figure 2 = 12 (1); Total = 19 (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 20$ (1); Total $= x + 2x + 20 = 3x + 20$ (1); $3x + 20 = 125$ so $3x = 105$ (1); $x = 35$ (1); Brother = ₹90 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 105 | Triangular numbers. |
| 2 | **(c)** $4n + 3$ | Check by substituting a value. |
| 3 | **(b)** 9 | Substitute carefully; xy is a product. |
| 4 | **(b)** 29 | Apply the rule. |
| 5 | **(b)** 32 | Square first, then multiply. |
| 6 | **(a)** 70 | Evaluate each bracket then multiply. |
| 7 | **(b)** $20x + 8$ | Perimeter $= 2(l + b)$. |
| 8 | **(c)** 210 | Triangular numbers. |
| 9 | **(c)** $10n + 5$ | Check by substituting a value. |
| 10 | **(d)** 6 | Substitute carefully; xy is a product. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $5n + 5(n + 1) + 5(n + 2)$ (1); $= 15n + 15 = 15(n + 1)$ (1). | 1+1 |
| 12 | Substitute: $3(4)^2 - 5(4) + 1$ (1); $= 29$ (1). | 1+1 |
| 13 | Expand: $8x - 8 - 5x + 20$ (1); $3x + 12$ (1). | 1+1 |
| 14 | $4n + 4(n + 1) + 4(n + 2)$ (1); $= 12n + 12 = 12(n + 1)$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $P = 2[(4x + 6) + (5x - 2)]$; $= 2[9x + 4]$ (1); $= 18x + 8$ (1); At $x = 8$: 152 cm (1). | 1+1+1 |
| 16 | $8n$ is even (even × any integer); Even + odd = odd (1); $n = 3$: 29 odd (1); $n = 4$: 37 odd (1). | 1+1+1 |
| 17 | A $= 10 + 18k$, B $= 70 + 8k$; Equate: $10k = 60$ (1); $k = 6$ km (1); Fare then ₹118 (1). | 1+1+1 |
| | **OR:** $P = 2[(6x + 2) + (4x - 3)]$; $= 2[10x + -1]$ (1); $= 20x + -2$ (1); At $x = 7$: 138 cm (1). | 1+1+1 |
| 18 | $10n$ is even (even × any integer); Even + odd = odd (1); $n = 4$: 47 odd (1); $n = 5$: 57 odd (1). | 1+1+1 |
| | **OR:** A $= 25 + 14k$, B $= 85 + 9k$; Equate: $5k = 60$ (1); $k = 12$ km (1); Fare then ₹193 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $6(20) + 1 = 121$ (1); (ii) $6n + 1 = 151$ gives $6n = 150$ (1); $n = 25$ (1); (iii) Figure 1 $= 7$, figure 2 $= 13$ (1); Total $= 20$ (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 12$ (1); Total $= x + 2x + 12 = 3x + 12$ (1); $3x + 12 = 162$ so $3x = 150$ (1); $x = 50$ (1); Brother $=$ ₹112 (1). | 1+1+1+1+1 |
| 20 | (i) $3(30) + 7 = 97$ (1); (ii) $3n + 7 = 49$ gives $3n = 42$ (1); $n = 14$ (1); (iii) Figure 1 $= 10$, figure 2 $= 13$ (1); Total $= 23$ (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 30$ (1); Total $= x + 2x + 30 = 3x + 30$ (1); $3x + 30 = 105$ so $3x = 75$ (1); $x = 25$ (1); Brother $=$ ₹80 (1). | 1+1+1+1+1 |

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 43 | Apply the rule. |
| 2 | **(b)** 120 | Square first, then multiply. |
| 3 | **(c)** 112 | Evaluate each bracket then multiply. |
| 4 | **(b)** $26x + 2$ | Perimeter $= 2(l + b)$. |
| 5 | **(c)** 325 | Triangular numbers. |
| 6 | **(c)** $3n + 3$ | Check by substituting a value. |
| 7 | **(a)** 28 | Substitute carefully; xy is a product. |
| 8 | **(a)** 55 | Apply the rule. |
| 9 | **(a)** 30 | Substitute and simplify. |
| 10 | **(d)** 28 | Substitute and simplify. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Substitute: $4(3)^2 - 2(3) + 6$ (1); $= 36$ (1). | 1+1 |
| 12 | Expand: $6x - 24 - 2x + 14$ (1); $4x - 10$ (1). | 1+1 |
| 13 | $6n + 6(n + 1) + 6(n + 2)$ (1); $= 18n + 18 = 18(n + 1)$ (1). | 1+1 |
| 14 | Substitute: $5(2)^2 - 6(2) + 7$ (1); $= 15$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $P = 2[(8x + 5) + (3x - 2)]$; $= 2[11x + 3]$ (1); $= 22x + 6$ (1); At $x = 4$: 94 cm (1). | 1+1+1 |
| 16 | $12n$ is even (even × any integer); Even + odd = odd (1); $n = 6$: 81 odd (1); $n = 7$: 93 odd (1). | 1+1+1 |
| 17 | A $= 40 + 13k$, B $= 100 + 8k$; Equate: $5k = 60$ (1); $k = 12$ km (1); Fare then ₹196 (1). | 1+1+1 |
| | **OR:** $P = 2[(3x + 7) + (6x - 4)]$; $= 2[9x + 3]$ (1); $= 18x + 6$ (1); At $x = 9$: 168 cm (1). | 1+1+1 |
| 18 | $2n$ is even (even × any integer); Even + odd = odd (1); $n = 8$: 27 odd (1); $n = 9$: 29 odd (1). | 1+1+1 |
| | **OR:** A $= 15 + 20k$, B $= 75 + 10k$; Equate: $10k = 60$ (1); $k = 6$ km (1); Fare then ₹135 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $7(10) + 4 = 74$ (1); (ii) $7n + 4 = 116$ gives $7n = 112$ (1); $n = 16$ (1); (iii) Figure 1 $= 11$, figure 2 $= 18$ (1); Total $= 29$ (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 18$ (1); Total $= x + 2x + 18 = 3x + 18$ (1); $3x + 18 = 153$ so $3x = 135$ (1); $x = 45$ (1); Brother $=$ ₹108 (1). | 1+1+1+1+1 |
| 20 | (i) $8(9) + 5 = 77$ (1); (ii) $8n + 5 = 109$ gives $8n = 104$ (1); $n = 13$ (1); (iii) Figure 1 $= 13$, figure 2 $= 21$ (1); Total $= 34$ (1). | 1+1+1+1+1 |
| | **OR:** Brother $= 2x + 25$ (1); Total $= x + 2x + 25 = 3x + 25$ (1); $3x + 25 = 205$ so $3x = 180$ (1); $x = 60$ (1); Brother $=$ ₹145 (1). | 1+1+1+1+1 |

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